

Supply Chain Operations Analytics Report

Delivery Performance & Operational Risk Assessment

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Tools: Python, Pandas, Power BI, DAX

Executive Summary

This report analyzes 180,519 order line records representing 65,752 unique customer orders to evaluate delivery performance, quantify profit risk associated with delayed shipments, and identify opportunities to improve supply chain operations. Python was used for data preparation and validation, while Power BI was used to build an executive dashboard.

Business Problem

Late deliveries reduce operational efficiency and customer satisfaction. This project evaluates delivery performance, identifies operational drivers of delays, and recommends areas for improvement.

Project Objectives

- Measure delivery performance using operational KPIs.
- Analyze regional and shipping mode performance.
- Quantify global **Profit at Risk** associated with delayed shipments.
- Build an executive dashboard.
- Provide actionable recommendations.

Dataset Overview

180,519 order line records

65,752 unique orders

Global fulfillment data across multiple regions and product categories.

Each row represents an order line rather than a complete order.

Data Preparation

Removed unnecessary personally identifiable fields, converted date columns, created shipping variance and on-time delivery metrics, validated duplicate orders, and investigated discrepancies between the provided Late_delivery_risk field and a derived late-delivery indicator (late_delivery_flag).

Reorganized dataset into STAR schema, Implemented a deterministic MD5 cryptographic hashing function to generate stateless, reproducible surrogate keys for composite dimension tables, enforcing data warehouse idempotency.

Methodology

Data Cleaning → Exploratory Analysis → KPI Development → Dashboard Development.

- **Data Warehouse Modeling:** Migrated the unoptimized transaction file into a performance-optimized **Star Schema** (Fact table surrounded by decoupled Dimension tables) to facilitate highly efficient columnar storage and rapid DAX execution in Power BI.

Dashboard Overview

The dashboard summarizes on-time delivery, profit at risk, regional performance, shipping mode performance, monthly trends, and category-level financial exposure.

Key Findings

- **Geographic Bottlenecks:** Global On-Time Delivery Rate baseline is low at 43%. Regional performance varies significantly, with Western Europe and Central America driving the highest volume of Profit Risk due to late orders.
- **Premium Service Erosion:** Second Class Shipping orders handled with same urgency as Standard, resulting in similar delivery delays and times between both modes. Profit Risk due to late shipments. The company faces premium service erosion and customer churn risk due to the nearly identical transit delays and process cycle times between the two shipping modes.
- Among all categories, **Fishing** represents the highest vulnerability, accounting for 19% (\$427K) of total global Profit at Risk due to late shipments, followed by **Cleats** (12%, \$280K), and **Camping & Hiking Gear** (10%, \$226K).

Recommendations

Prioritize operational improvements for low-performing shipping modes and regions (investigate warehouse operations regarding Second Class Shipping), monitor high-value categories with elevated profit at risk, and track standardized KPIs over time.

Limitations

On-Time In-Full (OTIF): This KPI was replaced with On-Time Delivery due to lack of data tracking order completeness. Assumptions were made that all shipments were 100% full and undamaged.

Carrier Level SLAs: Since third-party carrier SLAs are unavailable in the dataset, KPIs isolating vendor SLA compliance/performance or pertaining to root cause analysis of delivery volatility are not possible.

True Cost of Goods Sold (COGS): Unable to calculate fluctuating supply chain infrastructure costs, as well as local customs/tariffs due to lack of data.

Efficiency: Transit time efficiency or warehouse dwelling time is not given nor quantifiable with existing data. A linear order status audit log would be helpful in quantifying any potential bottlenecks.

Conclusion

This project demonstrates an end-to-end analytics workflow from raw data through executive reporting and provides actionable operational insights.